Worked example- Deflection calculation for simply supported beam

Beam shown below spans 20 m and supports a live load of 14.8 kN/m in addition to self weight.

(a) Check the adequacy of the beam at transfer using the provisions of Cl. 8.1.6.2

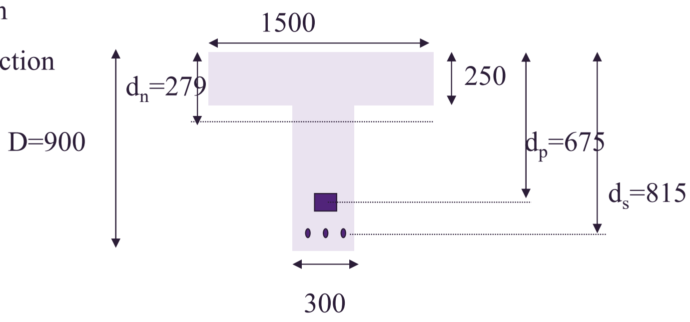
(b) Calculate the deflection of the beam.

-short term deflection at transfer

-short term deflection under service loads

-creep deflection

-long term deflection



f'_{cp} = 35 MPa
f'_c = 45 MPa

P_e = 1590kN
P_i = 1749kN
A_p = 1580mm^2
A_s = 1350 mm^2
A_g = 570,000 mm^2
I_g = 3.48 x 10^{10} mm^4
E_c = 33800 MPa
E_p = 200000 MPa
$Z_{bot} = I_g / y_{bot}$
= 3.48 x 10^{10} /621
=56.04x10^{6}mm^3

Problem 7 – Serviceability design

The cross-section of a post-tensioned concrete beam spanning 16 m, simply supported, is shown in Figure 7.1. The beam is post-tensioned with a tendon, which has a parabolic profile, with end eccentricity zero and maximum eccentricity at midspan of 350 mm. The tendon comprises of 10 number 12.7 mm super grade low relaxation strands. Tendon is stressed to 70% of the breaking load. Properties of materials are:

Concrete:

$f_c' = 32\ MPa$

$f_{cp}' = 25\ MPa$ (at 14 days when the tendon is stressed)

specific weight of concrete $= 24\ kN/m^3$

Prestressing Strands:

Cross-sectional area of a single strand $= 100\ mm^2$

Yield strength of tendons, $f_{py} = 1640\ MPa$

Minimum breaking load of strands $= 184\ kN$

$E_p = 190,000\ MPa$

Ordinary Reinforcement:

$A_{sc} = 4N24 = 1800\ mm^2,\ d_{sc} = 50mm$

$A_{st} = 8N24 = 3600\ mm^2,\ d_{st} = 950\ mm$

Cover: $50\ mm, 12\ mm$ shear reinforcement

$E_p = 190,000\ MPa, \quad E_s = 200,000\ MPa$

Assume immediate losses in prestress force to be 5%, and deferred losses to be 20%.

If the beam is subjected to a factored service load of $50\ kN/m$ (including self-weight);

1. Check the adequacy of the beam at transfer using Cl. 8.1.6.2
2. Calculate the total short term deflection at transfer
3. Calculate the total short-term deflection of the cracked prestressed member at service loads.
4. Calculate the additional long term deflection.
5. Does the beam satisfy the deflection criterion of span/250?

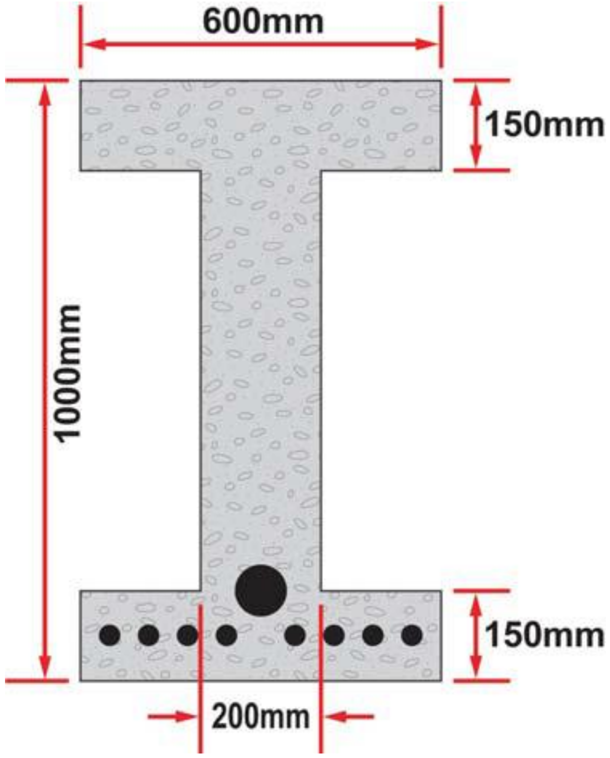


Figure 7.1 – Post-tensioned beam section